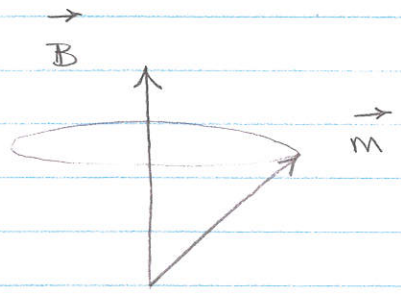
# Precessing Magnetic dipole radiation

Neda




magnetic dipole moment moves at an angular velocity $\omega$, find radiation?

$$\vec{\omega} = \frac{q}{2m}\vec{B}$$

we start with $\vec{A}(t) = \frac{\mu_0}{4\pi r c}\dot{\vec{m}}(t-\frac{r}{c}) \times \vec{n}$

$$\vec{B} = \vec{\nabla}\times\vec{A} = \frac{\mu_0}{4\pi c}\vec{\nabla}\times\left[\frac{1}{r}\dot{\vec{m}}(t-\frac{r}{c})\times\vec{n}\right]$$

$$= \frac{\mu_0}{4\pi c}\vec{\nabla}\times\left[\dot{\vec{m}}(t-\frac{r}{c})\times\frac{\vec{n}}{r}\right]$$

$$\vec{\nabla}\times(\vec{A}\times\vec{B}) = (\vec{B}\cdot\vec{\nabla})\vec{A} - (\vec{A}\cdot\vec{\nabla})\vec{B} + \vec{A}(\vec{\nabla}\cdot\vec{B}) - \vec{B}(\vec{\nabla}\cdot\vec{A})$$

$$\rightarrow \vec{B} = \frac{\mu_0}{4\pi c}\Big[(\frac{\vec{n}}{r}\cdot\vec{\nabla})\dot{\vec{m}}(t-\frac{r}{c}) - (\dot{\vec{m}}(t-\frac{r}{c})\cdot\vec{\nabla})\frac{\vec{n}}{r}$$
$$+ \dot{\vec{m}}(t-\frac{r}{c})(\vec{\nabla}\cdot\frac{\vec{n}}{r}) - \frac{\vec{n}}{r}(\vec{\nabla}\cdot\dot{\vec{m}}(t-\frac{r}{c}))\Big]$$

$$(\frac{\vec{n}}{r}\cdot\vec{\nabla})\dot{\vec{m}}(t-\frac{r}{c}) \sim \frac{n\ddot{m}}{rc} = \frac{\omega^2 m}{rc}$$

$$(\dot{\vec{m}}(t-\frac{r}{c})\cdot\vec{\nabla})\frac{\vec{n}}{r} \sim \frac{\dot{m}}{r^2} = \frac{\omega m}{r}$$

$$\rightarrow \frac{\text{Second term}}{\text{first term}} \sim \frac{\frac{\omega^2 m}{rc}}{\frac{\omega m}{r}} = \frac{\overset{\lambda}{c}}{r\,\omega} \sim \frac{\lambda}{r} << 1$$

So in far field we ignore derivitive of $\frac{\vec{n}}{r}$

$$\rightarrow \vec{\nabla} \times \left(\dot{\vec{m}}\left(t-\frac{r}{c}\right) \times \frac{\vec{n}}{r}\right) = \left(\frac{\vec{n}}{r} \cdot \vec{\nabla}\right) \dot{\vec{m}}\left(t-\frac{r}{c}\right) - \frac{\vec{n}}{r}\left(\vec{\nabla} \cdot \dot{\vec{m}}\left(t-\frac{r}{c}\right)\right)$$

$$\vec{\nabla} \cdot \vec{f} = \frac{\partial}{\partial r} \vec{f}\left(t-\frac{r}{c}\right) = \frac{\partial f}{\partial t} \cdot \frac{\partial t}{\partial r} = \frac{\partial \vec{f}}{\partial t} \nabla t = -\frac{\vec{n}}{c} \cdot \frac{\partial \vec{f}}{\partial t}$$

$$\vec{\nabla} = \frac{\partial}{\partial t'} \cdot \nabla t' = -\frac{\vec{n}}{c} \frac{\partial}{\partial t'}$$

$$\rightarrow \vec{B} = \frac{\mu_0}{4\pi c}\left[-\frac{1}{rc}\left((\vec{n} \cdot \vec{n})\ddot{\vec{m}}\left(t-\frac{r}{c}\right)\right) + \frac{\vec{n}}{rc}(\vec{n} \cdot \ddot{\vec{m}})\right]$$

$$\vec{B} = \frac{\mu_0}{4\pi c}\left[\frac{1}{rc}\left(\vec{n}(\vec{n} \cdot \ddot{\vec{m}}) - (\vec{n} \cdot \vec{n})\ddot{\vec{m}}\right)\right]$$

$$\vec{B} = \frac{\mu_0}{4\pi r c^2}\left(\vec{n} \times (\vec{n} \times \ddot{\vec{m}})\right)$$

Find $\vec{E}$ ? In far field : $\vec{E} = c\vec{B} \times \vec{n}$

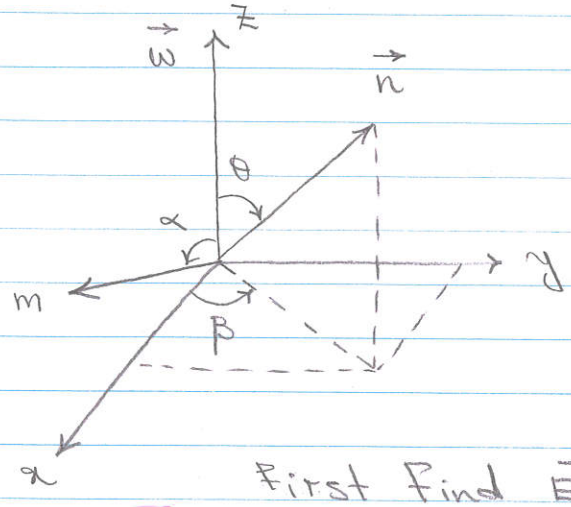
$$\vec{E} = \frac{\mu_0 c}{4\pi r c^2}\left(\vec{n} \times (\vec{n} \times \ddot{\vec{m}})\right) \times \vec{n}$$

$$= \frac{\mu_0}{4\pi r c}\left(\vec{n}(\vec{n} \cdot \ddot{\vec{m}}) - (\vec{n} \cdot \vec{n})\ddot{\vec{m}}\right) \times \vec{n}$$

$\vec{n} \cdot \ddot{\vec{m}}$ = scalar $\rightarrow$ (scalar) $\vec{n} \times \vec{n} = 0$

$\vec{n} \cdot \vec{n} = 1 \rightarrow (\vec{n} \cdot \vec{n})\ddot{\vec{m}} \times \vec{n} = \ddot{\vec{m}} \times \vec{n}$

$$\rightarrow \vec{E} = \frac{-\mu_0}{4\pi c r} \ddot{\vec{m}} \times \vec{n}$$




$\alpha$ is constant
m is rotating
for precessing m find
$E_\theta$ and $E_\varphi$ ?

First find $\vec{E}$ at point $\vec{n}$
and then we change coordinate to
spherical system

$$\vec{m} = m_0 \cos(\omega t) \sin\alpha \, \hat{x} + m_0 \sin(\omega t) \sin\alpha \, \hat{y} + m_0 \cos\alpha \, \hat{z}$$

$$\dot{\vec{m}} = m_0 \omega \left(-\sin(\omega t) \hat{x} + \cos(\omega t) \hat{y}\right) \sin\alpha$$

$$\ddot{\vec{m}} = -m_0 \omega^2 \left(\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y}\right) \sin\alpha$$

$$\vec{n} = \cos\theta \, \hat{z} + \sin\theta \sin\varphi \, \hat{y} + \sin\theta \cos\varphi \, \hat{x}$$

at $t' = 0$ : $\ddot{\vec{m}} = -m_0 \omega^2 \sin\alpha \, \hat{x}$

$$\vec{E} = -\frac{\mu_0}{4\pi c r} \ddot{\vec{m}} \times \vec{n}$$

$$\ddot{\vec{m}} \times \vec{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -m_0 \omega^2 \sin\alpha & 0 & 0 \\ \sin\theta \cos\varphi & \sin\theta \sin\varphi & \cos\theta \end{vmatrix} = \sin\theta \sin\varphi \, \hat{z} - \cos\theta \, \hat{y}$$

$$\rightarrow \vec{E} = \frac{\mu_0}{4\pi c r} m_0 \omega^2 \sin\alpha \left(\sin\theta \sin\phi \hat{z} - \cos\theta \hat{y}\right)$$

So we have $\vec{E}$ in Cartesian coordinate

$\vec{E}$ in spherical coordinate :

$$\vec{a} \equiv \sin\theta \sin\phi \hat{z} - \cos\theta \hat{y}$$

$$\vec{a} = a_\theta \hat{\theta} + a_\phi \hat{\phi}$$

$$a_\theta = -\hat{y} \cdot \hat{\theta} \cos\theta + \hat{z} \cdot \hat{\theta} \sin\theta \sin\phi$$

$$\hat{z} \cdot \hat{\theta} = -\sin\theta$$

$$\hat{y} \cdot \hat{\theta} = \cos\theta \sin\phi$$

$$\hat{x} \cdot \hat{\theta} = \cos\theta \cos\phi$$

$$a_\theta = -\cos^2\theta \sin\phi - \sin^2\theta \sin\phi = -\sin\phi$$

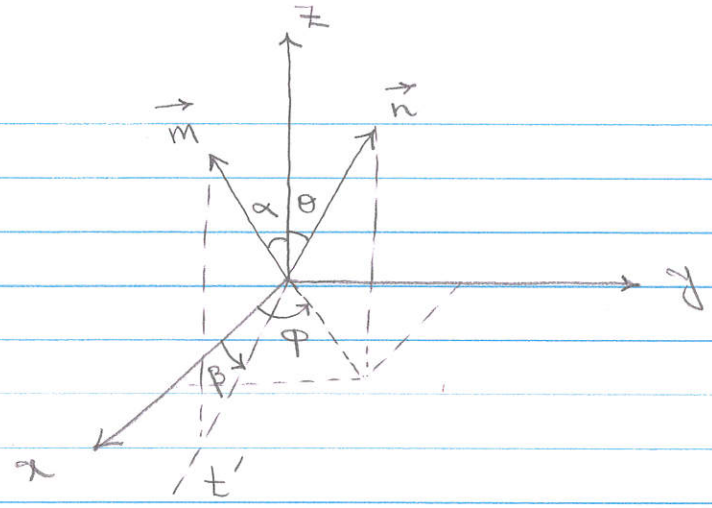
$$a_\phi = -\hat{y} \cdot \hat{\phi} \cos\theta + \hat{z} \cdot \hat{\phi} \sin\theta \sin\phi$$

$$\hat{y} \cdot \hat{\phi} = \cos\phi$$

$$\hat{z} \cdot \hat{\phi} = 0 \qquad (\phi \text{ is in the } x\text{-}y \text{ plane})$$

$$\rightarrow a_\phi = -\cos\theta \cos\phi$$

$$\vec{E} = \frac{\mu_0}{4\pi r c} m_0 \sin\alpha\, \omega^2 \left(-\sin\phi \hat{\theta} - \cos\phi \cos\theta \hat{\phi}\right)$$




Now consider a case that $\vec{m}$ rotates an angle $\beta$ in the direction of $\varphi$ :

So we simply replace $\varphi$ with $\varphi - \beta$

$\beta = \omega t'$ $\quad (t' = t - \frac{r}{c})$

$$\rightarrow \vec{E} = \frac{\mu_0}{4\pi r c} m_0 \sin\alpha\, \omega^2 \left(-\sin(\varphi - \omega t')\,\hat{\theta} - \cos(\varphi - \omega t')\cos\theta\,\hat{\varphi}\right)$$

if $\theta = 0 \rightarrow$ circularly polarized

$\theta = \frac{\pi}{2} \rightarrow$ linearly polarized

$\theta =$ arbitrary $\rightarrow$ elliptically polarized

Find intensity:

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$$

In Far Field: $\vec{B} = \frac{1}{c}\hat{n} \times \vec{E}$

$$\rightarrow \vec{S} = \frac{1}{\mu_0 c}(\vec{E} \times \hat{n} \times \vec{E})$$

$$\vec{S} = \frac{1}{\mu_0 c}\left[\hat{n}(\vec{E}\cdot\vec{E}) - \vec{E}\,\cancel{(\hat{n}\cdot\vec{E})}^{\,0}\right] = \frac{E^2}{\mu_0 c}\hat{n}$$

$$E^2 = \vec{E}\cdot\vec{E} = \left(\frac{\mu_0 m_0 \omega^2 \sin\alpha}{4\pi r c}\right)^2 \times$$

$$\left[\sin^2\left(\varphi - \omega\left(t - \frac{r}{c}\right)\right) - \cos^2\theta\cos^2\left(\varphi - \omega\left(t - \frac{r}{c}\right)\right)\right]$$

$$dI = r^2\,\vec{S}\cdot\hat{n}\,d\Omega \rightarrow \frac{dI}{d\Omega} = \vec{S}\cdot\hat{n}\,r^2$$

$$\left\langle \sin^2\left(\varphi - \omega\left(t - \frac{r}{c}\right)\right)\right\rangle = \frac{1}{2}$$

$$\left\langle \cos^2\left(\varphi - \omega\left(t - \frac{r}{c}\right)\right)\right\rangle = \frac{1}{2}$$

$$\rightarrow \overline{\frac{dI}{d\Omega}} = \left(\frac{\mu_0 m_0 \omega^2 \sin\alpha}{4\pi c}\right)^2 \frac{1}{2}(1 + \cos\theta)$$